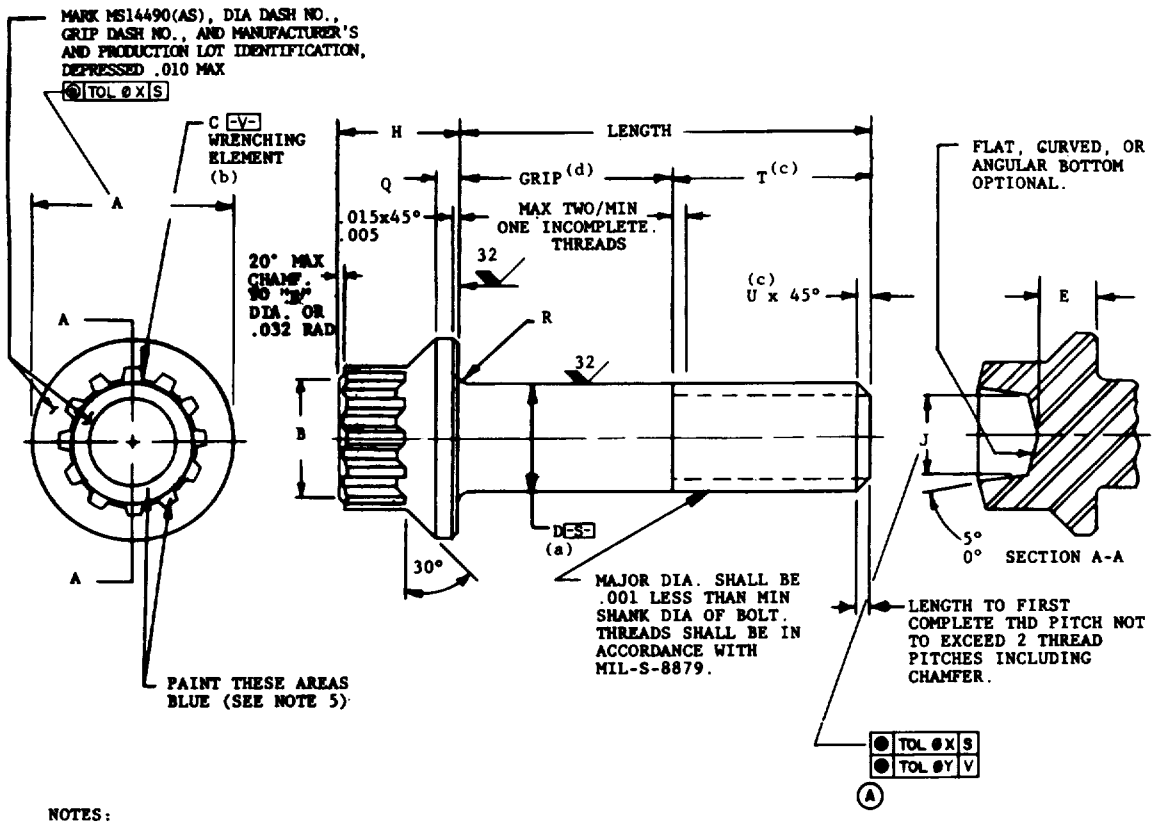
FED. SUP CLASS
5306



NOTES:

1. MATERIAL: INCONEL 718 PER AMS 5662, AMS 5663
2. HEAT TREATMENT: 220 KSI MIN AND 255 KSI MAX TENSILE STRENGTH AT ROOM TEMPERATURE
3. SURFACE TEXTURE: UNTHREADED SHANK AND BEARING SURFACE UNDER HEAD SHALL BE 32 MICROINCHES, ALL OTHER SURFACES SHALL BE 125 MICROINCHES IN ACCORDANCE WITH ANSI B46.1
4. FINISH: PASSIVATE PER QQ-P-35
5. PAINT: TOP OF BOLT WITH NO. 15102 BLUE PER TT-E-489
6. UNLESS OTHERWISE SPECIFIED, DIMENSIONS IN INCHES
   TOLERANCES: DECIMALS ±.010
   ANGLES ±2°
7. PERFORMANCE: PER MIL-B-008906(AS) EXCEPT FOR:
   (a) FATIGUE PER MIL-B-008906(AS) EXCEPT THAT LOADING SHALL BE AS SHOWN IN MS14490 TABLE II, AND MS14491 NUTS, MS14177 WASHERS SHALL BE USED; NO BOLT SHALL FAIL AT LESS THAN 2,000,000 CYCLES. THREADS AND BEARING SURFACE OF MS14490 BOLTS, MS14491 NUTS AND MS 14177 WASHERS SHALL BE LUBRICATED WITH MIL-T-5544 COMPOUND FOR THIS TEST.
   (b) ELEVATED TEMPERATURE TENSILE STRENGTH AND STRESS RUPTURE TESTS ARE NOT APPLICABLE.
8. PLATING: CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 2.
Ⓐ 9. ADMINISTRATIVE AND CONTRACTUAL PROVISIONS:

   a. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

   b. REFERENCED GOVERNMENT DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

Ⓐ DENOTES CHANGES

APPROVED 12 MAY 1986 REVISED Ⓐ 9 SEP 1986

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
This standard has been approved by the NAVAL AIR SYSTEMS COMMAND Department of the NAVY and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

P.A. NAVY – AS Other Cust	TITLE BOLT, WHEEL, INCONEL, 220 KSI Ftu, 132 KSI Fsu, 450°F EXTERNAL WRENCHING, SPLINE DRIVE, FLANGED HEAD.	MILITARY STANDARD MS14490(AS)
PROCUREMENT SPECIFICATION MIL-B-008906(AS)	SUPERSEDES:	SHEET 1 OF 3

DD FORM 672-1 (Limited coordination) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. PROJECT NO. 5306-N150 PLATE NO. 17709

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:44Z
Check the source to verify that this is the current version before use.

TABLE I

DIA. DASH NO.	NOMINAL SIZE	THREAD UNJF-3A (a)	A DIA.		B DIA.		C (b) WRENCHING ELEMENT	D DIA.		E MIN	H ±.010	J +.010 −.030	Q ±.010
			MAX	MIN	MAX	MIN		MAX	MIN				
04	1/4	.2500-28	.438	.428	.250	.235	10	.2495	.2485	.149	.250	.125	.045
05	5/16	.3125-24	.531	.521	.375	.360	12	.3120	.3110	.187	.312	.215	.061
06	3/8	.3750-24	.649	.639	.438	.423	14	.3745	.3735	.224	.337	.260	.069
07	7/16	.4375-20	.750	.740	.500	.485	16	.4370	.4360	.262	.393	.320	.082
08	1/2	.5000-20	.828	.818	.562	.547	20	.4995	.4985	.299	.450	.380	.102
09	9/16	.5625-18	.938	.928	.625	.610	22	.5615	.5605	.336	.505	.440	.112
10	5/8	.6250-18	1.050	1.040	.688	.673	24	.6240	.6230	.374	.562	.500	.123
12	3/4	.7500-16	1.230	1.220	.812	.797	28	.7490	.7480	.449	.600	.570	.155
14	7/8	.8750-14	1.438	1.428	.938	.923	32	.8740	.8730	.523	.700	.650	.180

TABLE II

DIA. DASH NO.	R RAD		T MIN REF (c)	U REF (c)	FATIGUE LOADING		TENSILE STRENGTH		DOUBLE SHEAR STRENGTH LB MIN	CONCENTRICITY		
					LOW (e) TENSION LOAD LB ±2%	HIGH (f) TENSION LOAD LB ±2%				X	Y	Z (a)
	MAX	MIN					MAX(LB)	MIN(LB)				
04	.041	.031	.560	.045	5,800	6,850	10,300	8,890	13,000	.005	.004	
05			.663	.052	9,200	10,900	16,400	14,100	20,200	.006	.005	.0045
06	.057	.047	.733		13,600	16,100	24,200	20,900	29,100	.008	.006	
07			.823	.063	18,400	21,800	32,800	28,300	39,700	.009	.008	
08			.903		24,600	29,100	43,800	37,800	51,800	.010	.008	
09			1.000	.069	31,100	36,500	55,500	47,900	65,600	.011	.009	.0060
10	.073	.063	1.060		38,900	43,100	69,500	59,900	81,000	.012	.010	
12			1.241	.078	56,500	62,600	100,800	86,900	117,000	.015	.012	
14			1.427	.089	77,400	85,700	137,000	119,000	159,000	.018	.014	.0090

NOTES:
(a) DIMENSIONS D AND THREAD PITCH DIAMETER SHALL BE CONCENTRIC TO EACH OTHER WITHIN VALUES SPECIFIED FOR Z T.I.R.
(b) WRENCHING ELEMENT PER MS33787. DRIVERS PER MIL-W-8982.
(c) REFERENCE DIMENSIONS ARE FOR DESIGN PURPOSES ONLY AND ARE NOT INSPECTION OR MANUFACTURING REQUIREMENTS. T = H MAX (MS14156) + 3 THDS + .153
(d) GRIP LENGTH OF BOLT SHALL BE MEASURED FROM UNDERSIDE OF HEAD TO THE END OF FULL CYLINDER PORTION OF SHANK.
(e) LOW TENSION FATIGUE LOAD IS BASED ON .65 x Ftu OF TABLE II.
(f) HIGH TENSION FATIGUE LOAD FOR SIZES 1/4 THROUGH 1/2 INCH ARE BASED ON .77 x Ftu AND SIZES 9/16 THROUGH 7/8 INCH ARE BASED ON .72 x Ftu OF TABLE II.

This standard has been approved by the NAVAL AIR SYSTEMS COMMAND, Department of the NAVY, and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

APPROVED 12 MAY 1988 REVISED (A) FOR CHANGES SEE PAGE 1

P.A. NAVY – AS Other Coast	TITLE BOLT, WHEEL, INCONEL, 220 KSI Ftu, 132 KSI Fsu, 450°F EXTERNAL WRENCHING, SPLINE DRIVE, FLANGED HEAD.	MILITARY STANDARD MS14490(AS)
PROCUREMENT SPECIFICATION MIL-B-008906(AS)	SUPERSEDES:	SHEET 2 OF 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:44Z
Check the source to verify that this is the current version before use.

TABLE III

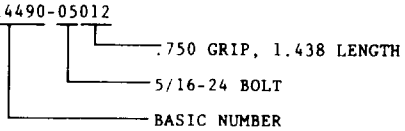
GRIP DASH NO.	GRIP ± .010	DIAMETER DASH NUMBER								
		04	05	06	07	08	09	10	12	14
		LENGTH ± .015								
004	.250	.835	.938	—	—	—	—	—	—	—
006	.375	.960	1.063	1.133	1.223	—	—	—	—	—
008	.500	1.085	1.188	1.258	1.348	1.428	—	—	—	—
010	.625	1.210	1.313	1.383	1.473	1.553	1.650	1.710	—	—
012	.750	1.335	1.438	1.508	1.598	1.678	1.775	1.835	2.016	—
014	.875	1.460	1.563	1.633	1.723	1.803	1.900	1.960	2.141	2.327
016	1.000	1.585	1.688	1.758	1.848	1.928	2.025	2.085	2.266	2.452
018	1.125	1.710	1.813	1.883	1.973	2.053	2.150	2.210	2.391	2.577
020	1.250	1.835	1.938	2.008	2.098	2.178	2.275	2.335	2.516	2.702
022	1.375	1.960	2.063	2.133	2.223	2.303	2.400	2.460	2.641	2.827
024	1.500	2.085	2.188	2.258	2.348	2.428	2.525	2.585	2.766	2.952
026	1.625	2.210	2.313	2.383	2.473	2.553	2.650	2.710	2.891	3.077
028	1.750	2.335	2.438	2.508	2.598	2.678	2.775	2.835	3.016	3.202
030	1.875	2.460	2.563	2.633	2.723	2.803	2.900	2.960	3.141	3.327
032	2.000	2.585	2.688	2.758	2.848	2.928	3.025	3.085	3.266	3.452
034	2.125	2.710	2.813	2.883	2.973	3.053	3.150	3.210	3.391	3.577
036	2.250	2.835	2.938	3.008	3.098	3.178	3.275	3.335	3.516	3.702
038	2.375	2.960	3.063	3.133	3.223	3.303	3.400	3.460	3.641	3.827
040	2.500	3.085	3.188	3.258	3.348	3.428	3.525	3.585	3.766	3.952
042	2.625	3.210	3.313	3.383	3.473	3.553	3.650	3.710	3.891	4.077
044	2.750	3.335	3.438	3.508	3.598	3.678	3.775	3.835	4.016	4.202
046	2.875	3.460	3.563	3.633	3.723	3.803	3.900	3.960	4.141	4.327
048	3.000	3.585	3.688	3.758	3.848	3.928	4.025	4.085	4.266	4.452
050	3.125	3.710	3.813	3.883	3.973	4.053	4.150	4.210	4.391	4.577
052	3.250	3.835	3.938	4.008	4.098	4.178	4.275	4.335	4.516	4.702
054	3.375	3.960	4.063	4.133	4.223	4.303	4.400	4.460	4.641	4.827
056	3.500	4.085	4.188	4.258	4.348	4.428	4.525	4.585	4.766	4.952
058	3.625	4.210	4.313	4.383	4.473	4.553	4.650	4.710	4.891	5.077
060	3.750	4.335	4.438	4.508	4.598	4.678	4.775	4.835	5.016	5.202
062	3.875	4.460	4.563	4.633	4.723	4.803	4.900	4.960	5.141	5.327
064	4.000	4.585	4.688	4.758	4.848	4.928	5.025	5.085	5.266	5.452
066	4.125	4.710	4.813	4.883	4.973	5.053	5.150	5.210	5.391	5.577
068	4.250	4.835	4.938	5.008	5.098	5.178	5.275	5.335	5.516	5.702
070	4.375	4.960	5.063	5.133	5.223	5.303	5.400	5.460	5.641	5.827
072	4.500	5.085	5.188	5.258	5.348	5.428	5.525	5.585	5.766	5.952
074	4.625	5.210	5.313	5.383	5.473	5.553	5.650	5.710	5.891	6.077
076	4.750	5.335	5.438	5.508	5.598	5.678	5.775	5.835	6.016	6.202
078	4.875	5.460	5.563	5.633	5.723	5.803	5.900	5.960	6.141	6.327
080	5.000	5.558	5.688	5.758	5.848	5.928	6.025	6.085	6.266	6.452
082	5.125	5.710	5.813	5.883	5.973	6.053	6.150	6.210	6.391	6.577
084	5.250	5.835	5.938	6.008	6.098	6.178	6.275	6.335	6.516	6.702
086	5.375	5.960	6.063	6.133	6.223	6.303	6.400	6.460	6.641	6.827
088	5.500	6.085	6.188	6.258	6.348	6.428	6.525	6.585	6.766	6.952
090	5.625	6.210	6.313	6.383	6.473	6.553	6.650	6.710	6.891	7.077
092	5.750	6.335	6.438	6.508	6.598	6.678	6.775	6.835	7.016	7.202
094	5.875	6.460	6.563	6.633	6.723	6.803	6.900	6.960	7.141	7.327
096	6.000	6.585	6.688	6.758	6.848	6.928	7.025	7.085	7.266	7.452
098	6.125	6.710	6.813	6.883	6.973	7.053	7.150	7.210	7.391	7.577
100	6.250	6.835	6.938	7.008	7.098	7.178	7.275	7.335	7.516	7.702
102	6.375	6.960	7.063	7.133	7.223	7.303	7.400	7.460	7.641	7.827
104	6.500	7.085	7.188	7.258	7.348	7.428	7.525	7.585	7.766	7.952
106	6.625	7.210	7.313	7.383	7.473	7.553	7.650	7.710	7.891	8.077
108	6.750	7.335	7.438	7.508	7.598	7.678	7.775	7.835	8.016	8.202
110	6.875	7.460	7.563	7.633	7.723	7.803	7.900	7.960	8.141	8.327
112	7.000	7.585	7.688	7.758	7.848	7.928	8.025	8.085	8.266	8.452
114	7.125	7.710	7.813	7.883	7.973	8.053	8.150	8.210	8.391	8.577
116	7.250	7.835	7.938	8.008	8.098	8.178	8.275	8.335	8.516	8.702
118	7.375	7.960	8.063	8.133	8.223	8.303	8.400	8.460	8.641	8.827
120	7.500	8.085	8.188	8.258	8.348	8.428	8.525	8.585	8.766	8.952
122	7.625	8.210	8.313	8.383	8.473	8.553	8.650	8.710	8.891	9.077
124	7.750	8.335	8.438	8.508	8.598	8.678	8.775	8.835	9.016	9.202
126	7.875	8.460	8.563	8.633	8.723	8.803	8.900	8.960	9.141	9.327

DIAMETER DASH NO. AT TOP OF TABLE, PLUS GRIP DASH NO. CONSTITUTES A COMPLETE PART NO.

FIRST TWO DASH NUMBERS DESIGNATE DIAMETER IN SIXTEENTHS. LAST THREE DASH NUMBERS DESIGNATE GRIP LENGTH IN SIXTEENTHS.

UNASSIGNED DASH NUMBERS SHALL NOT BE USED.

EXAMPLE OF PART NUMBER: MS14490-05012



APPROVED 12 MAY 1986 REVISED (A) FOR CHANGES SEE PAGE 1

This standard has been approved by the NAVAL AIR SYSTEMS COMMAND Department of the NAVY and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

P.A. NAVY - AS Other Cust	TITLE: BOLT, WHEEL, INCONEL, 220 KSI Ftu, 132 KSI Fsu, 450°F EXTERNAL WRENCHING, SPLINE DRIVE, FLANGED HEAD.	MILITARY STANDARD MS14490(AS)
PROCUREMENT SPECIFICATION MIL-B-008906(AS)	SUPERSEDES:	SHEET 3 OF 3

DD FORM SEP 87 672-1 (Limited coordination)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PLATE NO. 17709

☆U.S. GOVERNMENT PRINTING OFFICE: 1986 — 704-036/45474

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T13:44Z
Check the source to verify that this is the current version before use.